Interactive Display

CMPSC 105 – Data Exploration

Allegheny College

Running the Python3 Shell

- Type *python3* in the terminal to start the Python shell.

- Type statements or expressions at prompt:
    - `print("Hello, world")`
    - `x = 12**2`
    - `print(x)`
    - `print(x/2)`
    - `# bla bla bla...`
    - `# (This is a comment: everything after the # is ignored )`

- To exit the Python shell, type: exit() or Ctrl + Z to exit.

Data Types

- Integers, counting numbers
  - num_int = 1
- Floats, decimals
  - num_float = 3.1415
- Strings:
  - s_str = "Hello World”

Data Types

```python
height_int = 5
print(f" The height is: {height_int}")
print(" The height is:", height_int) # print another way

num_float = 3.14
print(f" The float variable is : {num_float}")

s_str = "Hello World"

print(" The integer is equal to: ", s_str)
```

Key Components

All programs built out of

- **Function calls**: Granting temporary kernel-time and/or using issuing parameters to a sub-sequence of instruction in a program.
- **Assignment statements**: The issuing of a value to a variable or place in memory to contain the value.
- **Iteration constructs**: Structures used in computer programming to repeat the same computer code multiple times (*loops*).
- **Conditional logic**: the use of logical rules in code to govern steps taken.
- **Variable creation**: The introduction of an object in memory to contain some value.
- **Variable computations**: The use of values contained in variables to create new value using an operator.
- **Variable output**: The revealing of some value in a variable by printing or another means.

Key Components

- **Function calls**: Granting temporary kernel-time and/or using issuing parameters to a sub-sequence of instruction in a program.

```
def greet(name):
    print(f"Hello, {name}! Welcome to Python.")

greet("Alice")
```

- **Assignment statements**: The issuing of a value to a variable or place in memory to contain the value.

```
x = 10  # First assignment (also creates x)
x = x + 5  # Reassigning a new value to x
y = x * 2  # Assigning a computed value to y
```

Key Components

- **Iteration constructs**: Structures used in computer programming to repeat the same computer code multiple times (*loops*).

```
fruits = ["apple", "banana", "cherry"]

for fruit in fruits:
    print(fruit)
```

- **Conditional logic**: the use of logical rules in code to govern steps taken.

```
age = 18

if age >= 7:
    print("You are an adult.")
```

Key Components

- **Variable creation: The introduction of an object in memory to contain some value**

```
x = 10  # 'x' is created and assigned the value 10
name = "Alice"  # 'name' is created and assigned a string value
```

- **Variable computations**: The use of values contained in variables to create new value using an operator.

```
a = 10
b = 5
sum_result = a + b       # Addition
diff_result = a - b      # Subtraction
prod_result = a * b      # Multiplication
quotient_result = a / b  # Division (float)
mod_result = a % b       # Modulus (remainder)
exp_result = a ** b      # Exponentiation (10^5)
```

Key Components

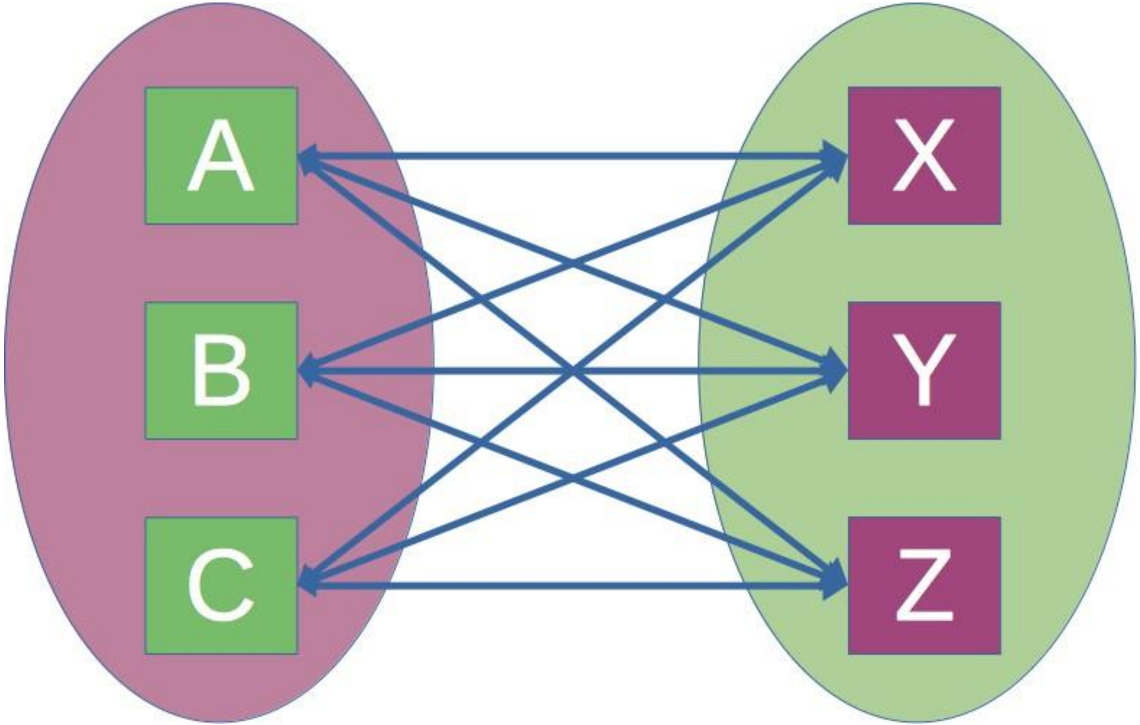
- **Variable output:** The revealing of some value in a variable by printing or another means

```
name = "Alice"
age = 25

print(name)  # Output: Alice
print(age)   # Output: 25
```

Mathematical Terminology

Is this a function?



NumPy

- NumPy is the foundational package for numerical computing in Python.
- Fast mathematical operations
- It works seamlessly with libraries such as pandas, matplotlib, and scikit-learn.

Why NumPy

```
# Python list
L = [1, 2, 3]
print(L * 2)
```

Output: [1, 2, 3, 1, 2, 3]

Why NumPy

```
import numpy as np

a = np.array([1, 2, 3])
print(a * 2)
```

Output: [2 4 6]

ipywidgets

- The interact function automatically creates sliders and controls for functions.
  - Make notebooks interactive
  - Support data visualization exploration